CLAIMS

1. An *in vitro* theranostic method of determining if a cancer patient is likely to be a good responder to an immune checkpoint inhibitor (ICI)-based therapy, comprising measuring, in a sample from said patient, the relative abundance of bacteria of the *Akkermansia* genus, wherein ~~the presence of~~when bacteria of the *Akkermansia* genus are present in said sample and their relative abundance in said sample is below a predetermined threshold comprised between 3 and 10%, ~~is indicative that~~ the patient is likely to be a good responder to the ICI-based therapy.

2. The method of claim 1, wherein the presence of bacteria of the *Akkermansia* genus above the predetermined threshold is indicative of dismal prognosis despite ICI-based therapy.

3. A method for determining if a cancer patient needs a bacterial compensation before administration of an ICI-based therapy, comprising measuring, in a sample from said patient, the relative abundance of bacteria of the *Akkermansia* genus, wherein:

(i) If no *Akkermansia* is present in the sample, the patient needs a bacterial compensation with at least bacteria of the *Akkermansia* genus before ICI administration;

(ii) if bacteria of the *Akkermansia* genus are present in the sample and their relative abundance in said sample is below a predetermined threshold comprised between 3 and 10%, the patient does not need any bacterial compensation before ICI administration; and

(iii) if the relative abundance of bacteria of the *Akkermansia* genus ~~are present~~ in the sample is above ~~a~~ the predetermined threshold, especially if this overrepresentation is consecutive to antibiotics exposure and/or associated with an overrespresentation of species belonging to the Gammaproteobacteria class and/or to the Desulfovibrionaceae family, the patient needs a bacterial compensation with a polymicrobial consortium and/or through fecal microbial transplant (FMT) from a healthy individual or from a cancer patient who successfully responded to the ICI-based therapy.

4. A method for determining if an individual has an intestinal microbiota dysbiosis, comprising measuring, in a sample from said individual, the relative

abundance of bacteria of the *Akkermansia* genus, wherein the presence of bacteria of the *Akkermansia* genus, with a relative abundance below a predetermined threshold comprised between 3 and 10%, is indicative that there is no intestinal microbiota dysbiosis.

5. The method of any of claims 1 to 4, wherein the relative abundance of *Akkermansia muciniphila* and/or *Akkermansia* SGB9228 is measured and compared to at least one predetermined threshold comprised between 3 and 10%.

6. The method of any of claims 1 to 5, wherein the predetermined threshold corresponds to a relative abundance between ~~1~~ 3 and ~~10~~6,5%.

7. The method of any of claims 1 to 6, wherein the predetermined threshold corresponds to a relative abundance between ~~3~~ 4,75 and 6.5%.

8. The method of any of claims 1 to 7, wherein the ICI-based therapy is an anti-PD1/PD-L1/PD-L2 Ab-based therapy.

9. The method of any of claims 1 to 8, wherein the patient suffers from non small cell lung cancer (NSCLC) or kidney cancer.

10. A method according to any of the preceding claims, wherein the sample from said patient or individual is a feces sample, a sample from the colon or ileal luminal content of said patient or individual or a mucosal biopsy from said patient or individual.

11. A fecal microbial composition for use in the treatment of a cancer patient having an overrepresentation of *Akkermansia* in his/her intestinal microbiota, wherein said composition originates from a healthy individual or from a cancer patient who successfully responded to the ICI-based therapy and is used in combination with an ICI-based therapy and wherein the patient has been identified as having an overrepresentation of *Akkermansia* in his/her intestinal microbiota by the method of claim ~~-~~3.

~~12. A bacterial composition comprising bacteria of the *Akkermansia* genus, for use in the treatment of a cancer patient having no *Akkermansia* in his/her intestinal microbiota, wherein said composition is used in combination with an ICI-based therapy.~~

~~13. The fecal microbial composition or the bacterial composition of claim 11 or claim 12, for the use of claim 11 or claim 12, wherein said bacteria of the *Akkermansia* genus are *Akkermansia* SGB9228.~~

~~14.~~12. The fecal microbial composition ~~or the bacterial composition of any one of claims 11 to 13~~according to claim 11, for the use of claim 11 ~~or claim 12~~, wherein the patient suffers from non small cell lung cancer (NSCLC) or kidney cancer.

~~15.~~13. The fecal microbial composition ~~or the bacterial composition~~ of ~~any one of~~ claim~~s~~ 11 ~~to 13~~or claim 12, for the use of claim 11 or claim 12, wherein the patient suffers from non-squamous NSCLC.